Software Requirements Specification

for

<Student Attendance System>

Version 1.0 approved

Prepared by Rajat R. Rokade

College

02-03-2026

Asst. Prof. Neha Nagdeve

Table of Contents

Table of Contents **2**
Revision History ..... Error! Bookmark not defined.
1. Introduction **1**
1.1 Purpose 1
1.2 Document Conventions ..... **Error! Bookmark not defined.**
1.3 Intended Audience and Reading Suggestions 1
1.4 Product Scope 1
1.5 References 1
2. Overall Description **2**
2.1 Product Perspective 2
2.2 Product Functions 2
2.3 User Classes and Characteristics 2
2.4 Operating Environment 2
2.5 Design and Implementation Constraints 3
2.6 User Documentation 3
2.7 Assumptions and Dependencies 3
3. External Interface Requirements **3**
3.1 User Interfaces 3
3.2 Hardware Interfaces 3
3.3 Software Interfaces 3
3.4 Communications Interfaces 3
4. System Features **4**
4.1 System Feature 1 4
4.2 System Feature 2 (and so on) ..... **Error! Bookmark not defined.**
5. Other Nonfunctional Requirements **4**
5.1 Performance Requirements 5
5.2 Safety Requirements 5
5.3 Security Requirements 5
5.4 Software Quality Attributes 5
5.5 Business Rules 5
6. Other Requirements **5**
Appendix A: Glossary **5**
Appendix B: Analysis Models **6**
Appendix C: To Be Determined List ..... Error! Bookmark not defined.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Student Attendance System (SAS), version 1.0. The SAS is an automated, web-based system designed to digitize and streamline the process of recording, tracking, and reporting student attendance across an academic institution.

1.2 Intended Audience and Reading Suggestions

- Project Manager
- Developers
- QA/Testers
- UI/UX designers
- System Administrator
- Faculty

1.3 Product Scope

The Student Attendance System helps institutions:

- Record attendance digitally
- Generate attendance reports
- Track student performance
- Reduce paperwork and errors

1.4 References

<List any other documents or Web addresses to which this SRS refers. These may include user interface style guides, contracts, standards, system requirements specifications, use case documents, or a vision and scope document. Provide enough information so that the reader could access a copy of each reference, including title, author, version number, date, and source or location.>

2. Overall Description

2.1 Product Perspective

- This is a standalone system that can also integrate with existing school/college management systems.

2.2 Product Functions

- Student registration
- Teacher login
- Mark attendance
- Generate reports
- View attendance history

2.3 User Classes and Characteristics

- **Admin**
  - Full access
  - Technical knowledge required
- **Teacher**
  - Marks attendance
  - Basic computer skills
- **Student**
  - Views attendance
  - Minimal technical skills

2.4 Operating Environment

- Windows/Linux/Andriod
- Chrome, Edge

2.5 Design and Implementation Constraints

- Must Use Secure login System
- Database required(Mysql/MongoDB)
- Internet Connection Required

2.6 User Documentation

- User Manual
- Online help system
- Tutorials

2.7 Assumptions and Dependencies

- Users have Internet Access
- Systems depend on database availability
- Accurate user input is assumed

3. External Interface Requirements

3.1 User Interfaces

- Login Page
- Dashboard
- Attendance Marking screen
- Reports PAge

3.2 Hardware Interfaces

- Computer/Mobile device
- Server System

- ### 3.3 Software Interfaces

- Database (MySql)
- Web Browser
- Backend Framework

3.4 Communications Interfaces

- HTTP/HTTPS protocol
- Internet Connectivity

4. System Features

4.1 User Authentication

- *Allows users to securely log in.*
- User enters login details → System validates → Access granted/denied
- *REQ-1: System shall allow user login*
- *REQ-2: System shall validate username and*

4.2 Attendance Tracking

- *Teachers mark attendance for students.*
- *Teacher selects class → Marks attendance → System saves data*
- *REQ-4: System shall allow attendance marking*
- *REQ-5: System shall store attendance records*
- *REQ-6: System shall prevent duplicate entries*

4.3 Report Generation

- Generate attendance reports.
- *User requests report → System generates report*
- REQ-7: System shall generate daily reports
- REQ-8: System shall generate monthly reports
- REQ-9: System shall allow report download

4.4 Student Dashboard

- *Students can view attendance status.*
- *Student logs in → Views attendance*
- REQ-10: System shall display attendance percentage
- REQ-11: System shall show attendance history

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- System should respond within 2 seconds
- Support multiple users simultaneously

5.2 Safety Requirements

- Prevent data loss
- Backup system required

5.3 Security Requirements

- Secure login authentication
- Data encryption
- Role-based access control

5.4 Software Quality Attributes

- Reliability
- Usability
- Maintainability
- *Scalability*

5.5 Business Rules

- Attendance must be marked daily
- Only teachers can mark attendance
- Admin manages users

6. Other Requirements

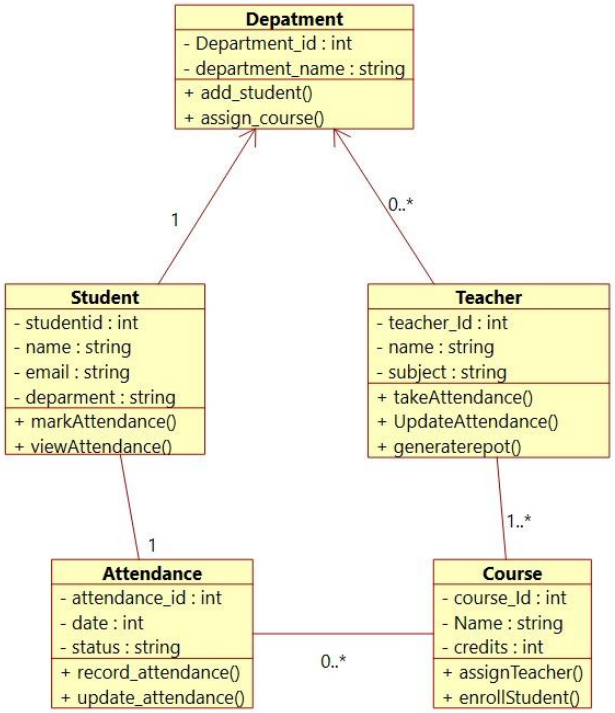
- Database design required
- Backup system implementation
- Legal compliance (data privacy)

Appendix A: Glossary

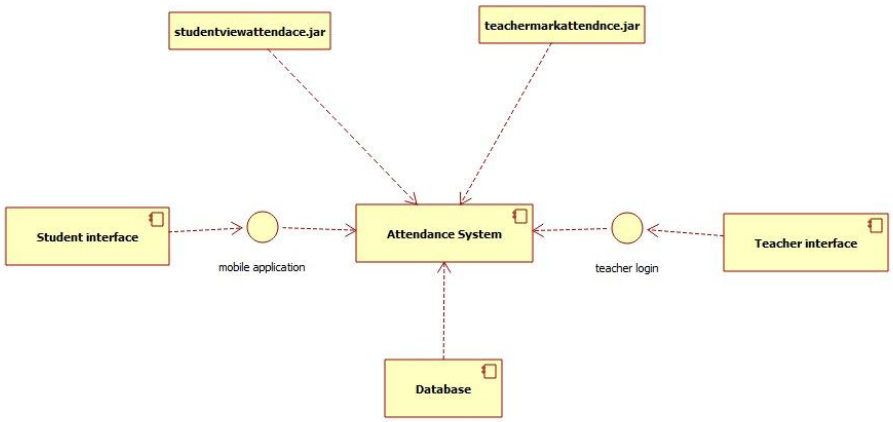
- SRS: Software Requirements Specification
- Admin: System administrator
- User: Teacher/Student

Appendix B: Analysis Models

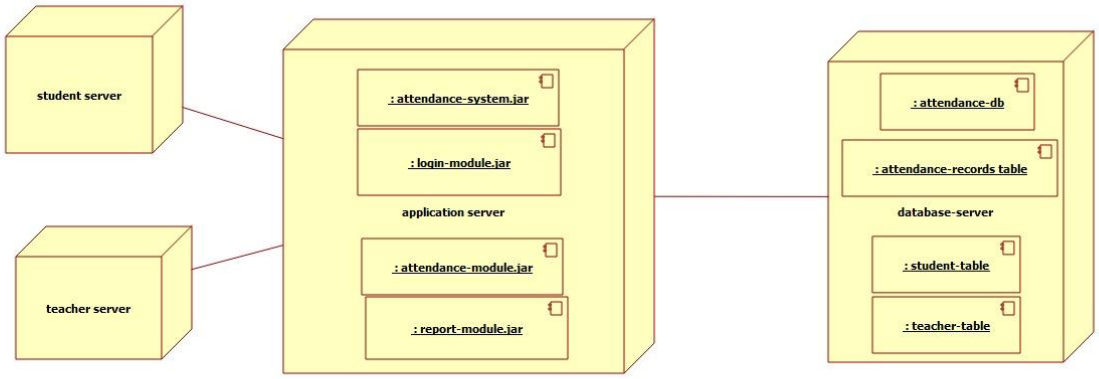
1. Class Diagram



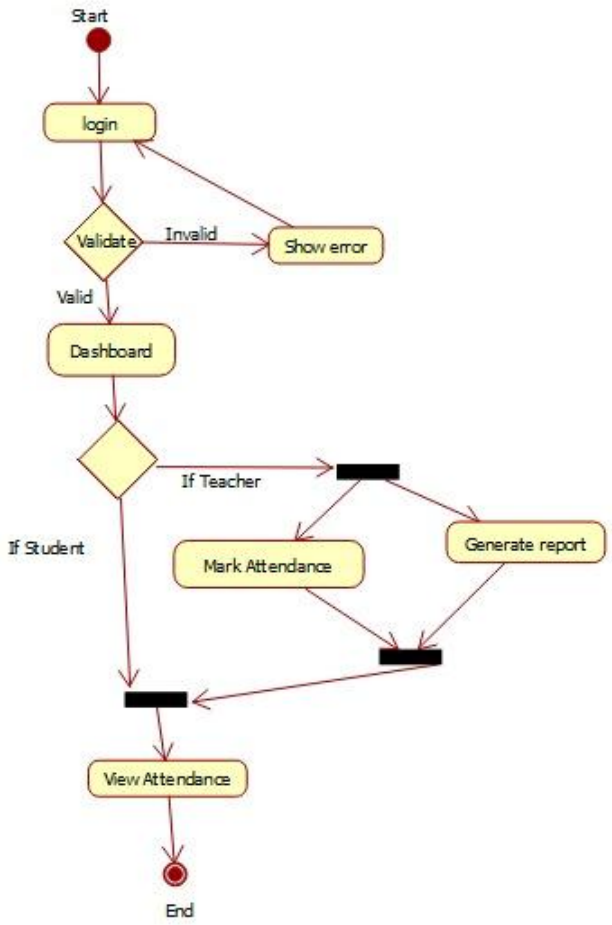
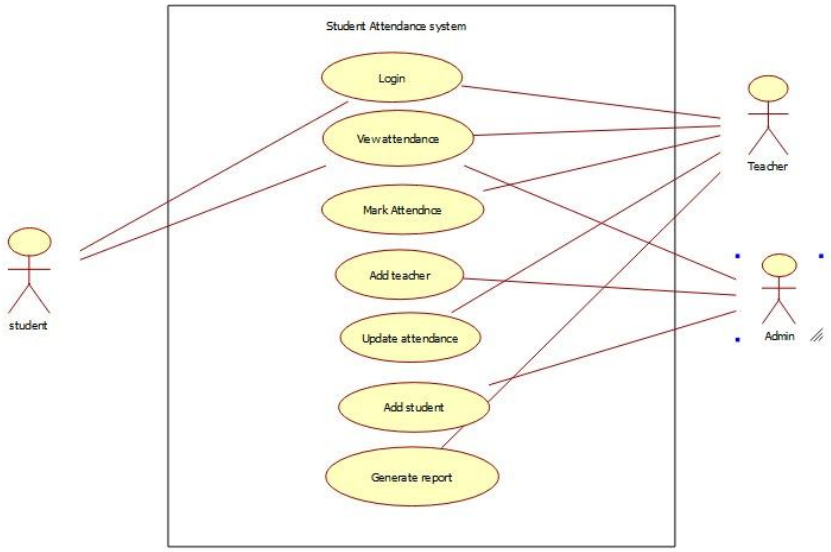
2. Component Diagram



3.Deployment Diagram



4.Activity Diagram



5.Use Case diagram